

Reporte de resultados

Prediction Models of Individual-Level Healthcare Costs after Moderate-to-Severe Traumatic Injury

Mari Storli Rasmussen*, PhD, Diego Rivera^{1,2*}, PhD, Laiene Olabarrieta-Landa^{1,2*}, PhD,

Håkon Øgreid Moksnes, MD, PhD, Christoph Schäfer, MD, PhD, Torgeir Hellstrøm, MD, PhD

Cecilie Røe, MD, PhD, Nada Andelic, MD, PhD

El objetivo de este estudio fue desarrollar modelos de predicción de los costes sanitarios a nivel individual asociados a la atención de TCE durante el primer año posterior a la lesión.

Gamma GLM

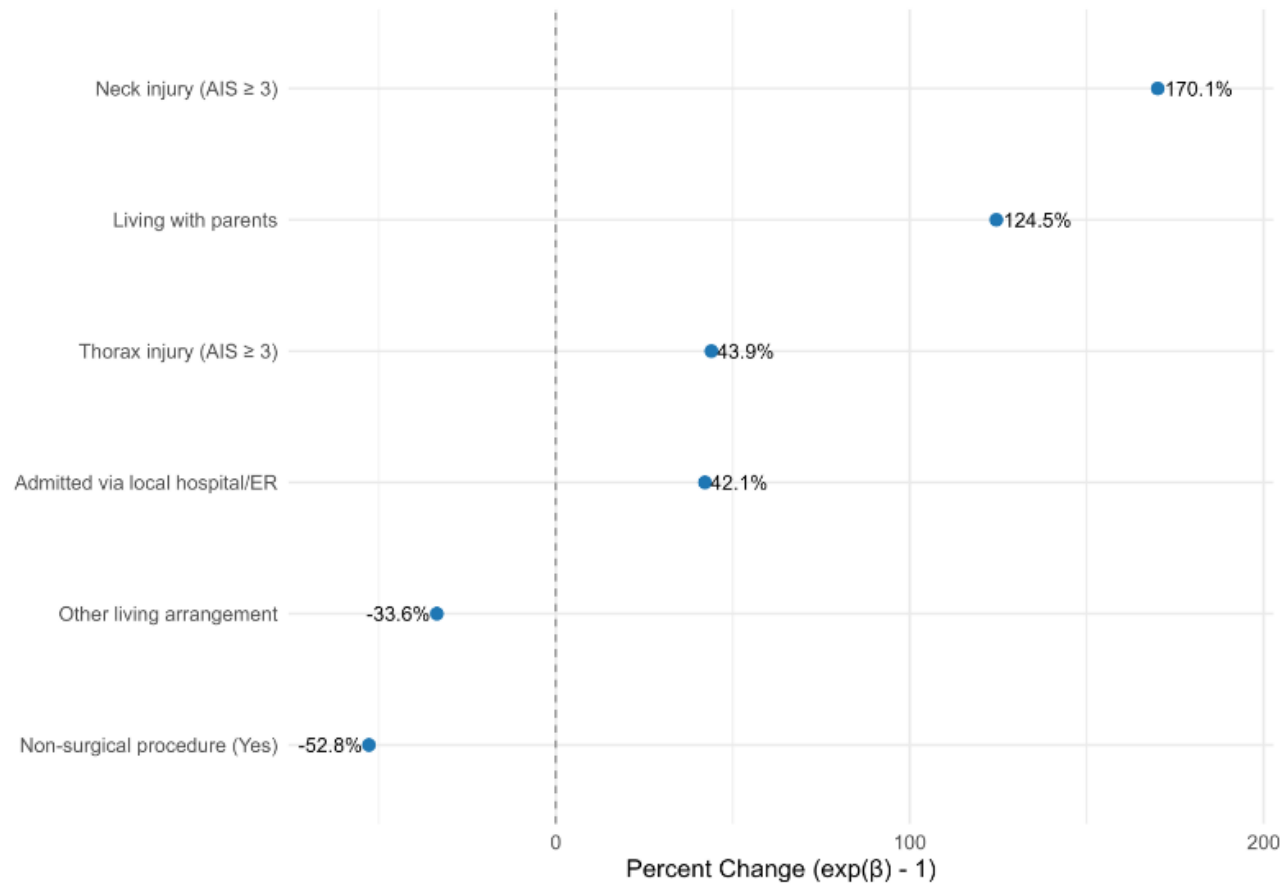
Annual costs (NOK) by service categories: coefficient estimates from Gamma (log-link) GAMs for primary care, secondary care, and total costs

Cost	Coefficients	Estimate	Std. Error	<i>t</i> value	<i>p</i> value
Primary Care	(Intercept)	8.584	0.147	58.222	<0.001
	Length of stay (acute hospital)	0.017	0.007	2.570	0.011
	Age (years)	0.005	0.002	2.297	0.022
	Number of injuries	0.021	0.013	1.639	0.102
	Neck injury (AIS ≥ 3)	0.906	0.313	2.895	0.004
	Discharged to local hospital	0.239	0.101	2.358	0.019
	Discharged to rehabilitation	0.274	0.126	2.176	0.030
	Discharged to nursing home/others	0.284	0.275	1.030	0.304
	TBI (Yes)	0.130	0.092	1.409	0.160

In primary care costs, the most relevant predictors were presence of severe neck injury ($\beta = 0.906$), discharge to nursing home/others ($\beta = 0.284$), to rehabilitation ($\beta = 0.274$), and to local hospital ($\beta = 0.239$), presence of traumatic brain injury ($\beta = 0.130$), number of injuries ($\beta = 0.021$), acute hospital length of stay ($\beta = 0.017$), and age ($\beta = 0.005$).

B

Estimated % Change: Specialist Cost



$$(\% \text{ Change}_j = [e^{\beta_j} - 1] \cdot 100)$$

- Positive specialist service costs were modeled using a gamma-log specification. Neck injuries ($\beta = 1.527$) and being under parental care ($\beta = 0.695$) were associated with substantially higher costs, corresponding to increases of 170.1% and 124.5%, respectively. Abdominal ($\beta = 0.634$) and thorax injuries ($\beta = 0.409$) also led to higher expected expenditures. In contrast, other living arrangement, referring to participants who were not married or cohabitating and who did not live with their parents, ($\beta = -0.580$) and higher number of injuries ($\beta = -0.060$) were linked to cost reductions of 33.6% and 52.8%, respectively.

Normative data for verbal fluency and naming tests in Spanish-speaking adults in the United States

Laiene Olabarrieta-Landa^{a,b}, Diego Rivera^{a,b}, Bridget Xia^c, Oswaldo Moreno^d, Daniela Ramos Usuga^e, Gloria M. Morel Valdés^f, Enrique Lopez^{g,h}, Miriam J. Rodriguez^{i,j}, Carmen I. Carrión^k, Anthony H. Lequerica^{l,m}, Christin I. Dragoⁿ, Patricia García^{o,p}, Patricia M. Rivera^q, Paul B. Perrin^{c,r} and Juan Carlos Arango-Lasprilla^{d,*}

El objetivo de este estudio fue establecer datos normativos para pruebas de fluidez verbal y denominación en una muestra de hablantes hispanos residentes en distintas regiones de Estados Unidos.

Fluidez verbal

- Se trata de generar el mayor número de palabras posibles en un tiempo determinado (generalmente 60 segundos)
 - Son de fácil administración y muy rápidos
 - Altamente sensibles al daño cerebral

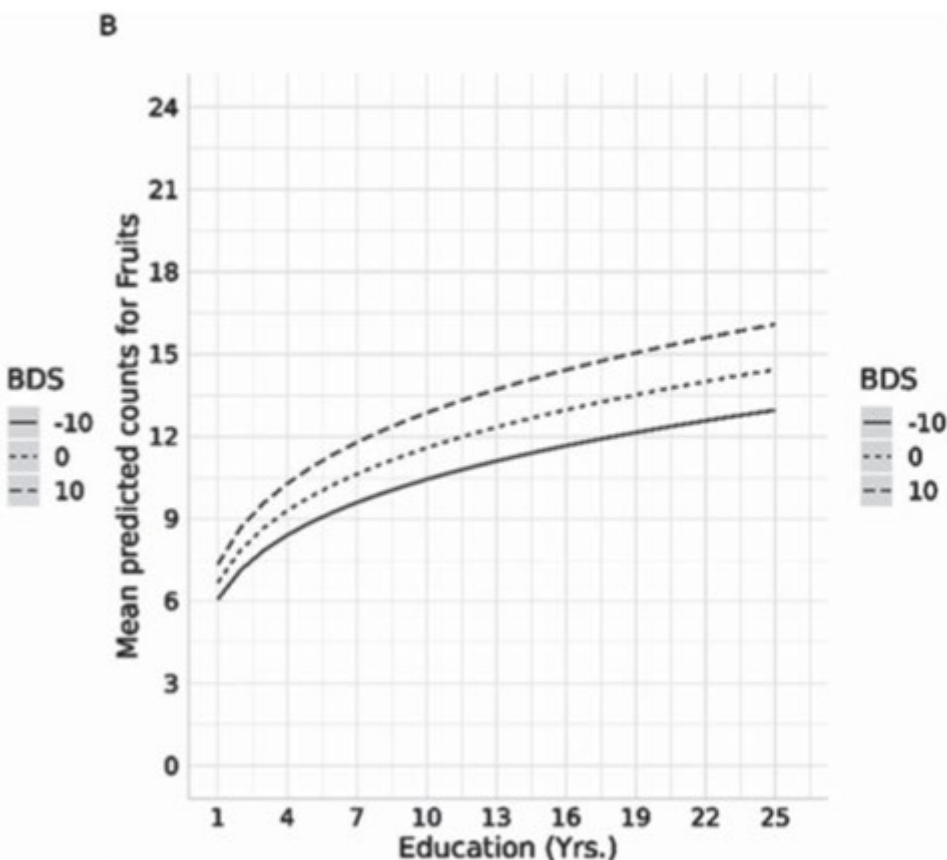
¿Tipo de distribución?

Poisson



Mean, median, mode, and credibility interval (95%) for the posterior distributions of the parameters in the semantic verbal fluency and Boston Naming Test models

Test score	Parameters	Mean	Median	Mode	HDI Low	HDI High
Animals	Intercept	2.1931	2.2003	2.2199	1.7833	2.6111
	<i>log(education)</i>	0.2267	0.225	0.219	0.0799	0.3782
	BDS	−0.0132	−0.0134	−0.0133	−0.0368	0.0105
	<i>log(education)*BDS</i>	0.0089	0.0089	0.0087	5.00×10^{-4}	0.018
Fruits	Intercept	1.8868	1.8857	1.8779	1.45	2.3161
	<i>log(education)</i>	0.2438	0.244	0.239	0.0723	0.3921
	BDS	0.0098	0.0096	0.0082	−0.0157	0.0342
	<i>log(education)*BDS</i>	3.00×10^{-4}	4.00×10^{-4}	0.0012	−0.0091	0.009



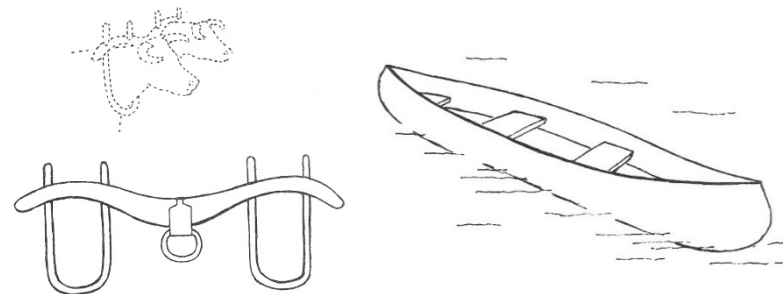
For Animal and Fruit categories, an education by BDS interaction was found, such that after 5 years of education, people with higher education and greater Spanish proficiency produced a greater number of animals than those with greater English proficiency. However, in fruits category, the differences between Spanish and English proficiency were maintained regardless of educational attainment, and people with greater Spanish proficiency always produced a greater number of fruits (Fig. 2A, B). See Table 4 for mean, median, mode, and credibility interval (95%) for the posterior distributions for the of the parameters for each model.

Test de denominación de Boston

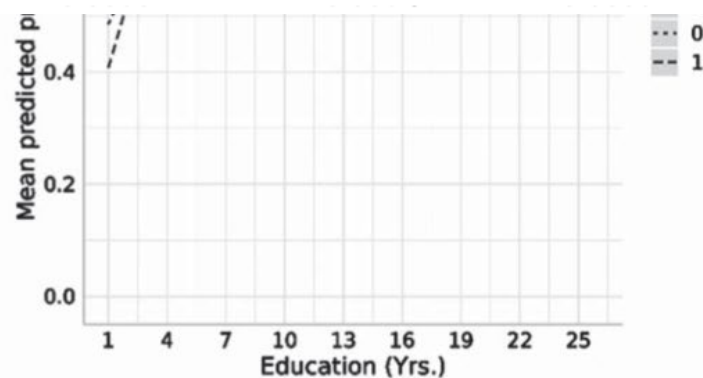
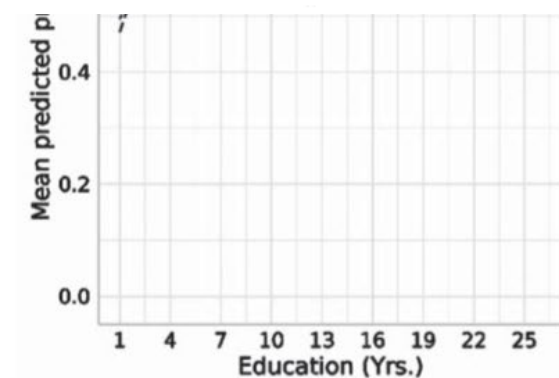
- Batería de evaluación de la afasia y de trastornos relacionados (Goodglass y Kaplan, 1996)
- **Tarea:** denominar una serie de láminas en blanco y negro presentadas en orden de dificultad
- **Versiones:** 60, 30, 15 y 11 ítems

¿Tipo de distribución?

Binomial



Test score	Parameters	Mean	Median	Mode	HDI Low	HDI High
BNT	Intercept	0.2112	0.2989	0.3079	-2.0554	1.8876
	$\log(\text{education})$	0.1599	0.1226	0.1026	-0.4931	0.9815
	BAS-Hispanic	-0.0792	-0.106	-0.0805	-0.6112	0.5394
	z-BDS	-0.319	-0.3171	-0.3315	-0.6547	0.0281
	$\log(\text{education}) * \text{BAS-Hispanic}$	0.0879	0.0994	0.103	-0.1491	0.2818
	$\log(\text{education}) * \text{z-BDS}$	0.2464	0.2451	0.2412	0.1143	0.369



3.5. Boston Naming Test (BNT)

Education was related to the BNT total score, such that scores increased according to the natural logarithmic transformation of education. In addition, an interaction between education by BAS and education by BDS were found, such that people with higher Hispanic acculturation and greater Spanish proficiency obtained higher scores than those with lower Hispanic acculturation and greater English proficiency. Moreover, performance increased with greater years of education (Table 4 and Fig. 2C, D).

Analysis of sociodemographic and linguistic influences on verbal fluency performance in Basque and Catalan bilingual adults

Laiene Olabarrieta-Landa^{1,2,*},, Diego Rivera^{1,2},, Montserrat Alegret^{3,4}, Alba Pérez³, Anabel Forte⁵, and Juan Carlos Arango-Lasprilla^{6,7},

El objetivo fue generar datos normativos para pruebas de fluidez verbal en hablantes de euskera y catalán de España.

- Se calcularon tres GLMM, uno para fluidez verbal fonológica/letras, uno para fluidez verbal semántica y uno para fluidez verbal de verbos, en cada grupo lingüístico.
- Los modelos completos incluyeron los siguientes predictores:
 - Idioma
 - Español codificado como 0 frente a lengua regional codificada como 1
 - Edad
 - Edad²
 - Educación formal
 - Educación transformada logarítmicamente
 - Sexo
 - Tipo de tarea (letras vs categorías)
 - Interacciones entre idioma y tipo de tarea

Parameter		Full model				Model 2			
		Mean	Median	HDI low	HDI high	Mean	Median	HDI low	HDI high
Phonologic	Intercept	0.885	0.896	0.421	1.406	0.928	0.924	0.492	1.370
	Language (Basque)	-0.056	-0.056	-0.128	0.009	-0.056	-0.056	-0.123	0.009
	Age	-0.339	-0.342	-1.090	0.414	-0.280	-0.284	-1.026	0.500
	Age ²	-0.856	-0.855	-1.603	-0.071	-0.870	-0.865	-1.606	-0.061
	log(Education)	0.592	0.587	0.413	0.768	0.590	0.592	0.416	0.754
	Sex (Men)	0.027	0.027	-0.062	0.121	-	-	-	-
	Letter B	-0.184	-0.184	-0.258	-0.118	-0.184	-0.183	-0.250	-0.106
	Letter E	-0.046	-0.045	-0.112	0.020	-0.047	-0.048	-0.118	0.020
	Letter F	0.087	0.088	0.024	0.152	0.088	0.087	0.018	0.152
	Letter M	0.193	0.193	0.133	0.257	0.192	0.192	0.132	0.260
	Letter P	0.277	0.277	0.217	0.340	0.277	0.276	0.215	0.337
	Letter R	0.178	0.178	0.112	0.240	0.177	0.176	0.113	0.238
	Letter S	0.157	0.158	0.092	0.221	0.157	0.157	0.087	0.217
	Language * Letter B	0.293	0.293	0.196	0.388	0.293	0.293	0.200	0.391
	Language * Letter E	0.051	0.051	-0.050	0.146	0.052	0.052	-0.046	0.143
	Language * Letter F	-0.868	-0.869	-0.973	-0.763	-0.868	-0.869	-0.974	-0.748
	Language * Letter M	-0.277	-0.278	-0.369	-0.183	-0.277	-0.276	-0.369	-0.181
	Language * Letter P	-0.443	-0.443	-0.536	-0.350	-0.443	-0.443	-0.542	-0.352

$$\log(\mu_{ij}) = \beta_0 + b_{0i} + \beta_1 X_{1ij} + \cdots + \beta_k X_{kij},$$

[Calculadora](#)

$$\begin{aligned}
 \lambda &= \exp(2.154 + (-0.191 \cdot 0) + (-0.881 \cdot 0.0320) + (-1.172 \cdot -0.0473) + (0.314 \cdot 2.4849) + (0.017 \cdot 0)) \\
 &= \exp(2.154 - 0.0282 + 0.0554 + 0.7802) \\
 &= \exp(2.917204) = 18.49.
 \end{aligned}$$

Estándares internacionales

- No existe una única guía
- El reporte debe ajustarse al tipo de estudio y al objetivo del análisis

Contexto	Guía
Investigación cuantitativa en psicología	APA Journal Article Reporting Standards – JARS
Investigación empírica en educación	AERA Standards for Reporting on Empirical Social Science Research
Ensayos controlados aleatorizados de intervenciones sociales y psicológicas	CONSORT-SPI
Estudios de caso	CARE Guidelines
Revisiones	PRISMA

- [Tablas y gráficas](#)

<https://www.equator-network.org/>